Wildfire Risk Prediction and Insurance Premium Modeling Using Hybrid Quantum-Classical Machine Learning

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GitHub Repository: `https://github.com/Akhil-Kambhatla/deloitte-qsc-2026`

Platform: IBM Qiskit 2.3.1, Qiskit Aer statevector simulator

Abstract

We present a hybrid quantum-classical machine learning pipeline for predicting wildfire risk across 2,593 California zip codes and modeling insurance premiums under fire exposure. Our approach addresses all three challenge tasks through a unified framework that connects wildfire science, quantum circuit design, and financial risk modeling.

For Task 1A, we engineered 43 features spanning four domains-fire history, weather and moisture, insurance exposure, and socioeconomic indicators-aggregated at the zip-code-year level from 2018 to 2022. We developed five distinct 10-qubit variational quantum classifier (VQC) configurations, including a novel contribution: reorganizing qubit assignments by domain cluster rather than by feature importance rank. This reordering improved the quantum model's ability to capture known 2023 fire locations from 37% to 63%, a 70% improvement without changing the circuit architecture. We additionally designed a custom "WildfireQCircuit" whose entanglement topology reflects known physical interactions from wildfire literature-moisture-fuel coupling, insurance-exposure feedback, and demographic vulnerability amplification-and conducted a controlled ablation study comparing domain-informed, generic, and random entanglement patterns.

For Task 1B, we provide an honest evaluation: our best quantum model (F1=0.337) reaches 66% of the best classical XGBoost performance (F1=0.509). We identify the NISQ-era optimization bottleneck as the primary limiting factor and demonstrate through ablation that circuit topology choices measurably affect prediction quality, even when they do not yet surpass classical baselines.

For Task 2, we built a time-series regression model predicting 2021 earned premiums using lag features, fire claims history, and wildfire risk scores from Task 1. Our XGBoost regressor achieved $R^2 = 0.881$ with a median absolute percentage error of 6.0%, substantially outperforming a naive last-year baseline ($R^2 = 0.735$). The quantum model's fire risk predictions were integrated as an input feature, completing the pipeline from wildfire risk assessment to financial impact.

Our central finding is that *how* features are mapped to qubits matters as much as *which* features are selected. This suggests that domain-informed quantum circuit design-translating scientific knowledge into entanglement topology-represents a principled path toward quantum utility in environmental risk modeling, even within current hardware constraints.

All code, data pipelines, and trained model outputs are available in our public GitHub repository.

Introduction

California's wildfire crisis has intensified dramatically in recent years. Nine of the state's ten warmest years on record have occurred since 2014, accelerating drought cycles, vegetation drying, and fire spread [1]. The 2024 Los Angeles fire season alone produced over 8,000 wildfires across more than 1 million acres, underscoring the urgent need for improved risk modeling. Rising temperatures interact with hydroclimate whiplash-rapid shifts between wet and dry conditions-to create heavier fuel loads that ignite more readily during prolonged dry seasons. Santa Ana and Diablo winds further compound the problem by pushing fires unpredictably into populated areas.

These cascading environmental dynamics create high-dimensional, non-linear interactions between climatic, ecological, and anthropogenic variables that challenge conventional modeling approaches [2, 3]. The fire triangle-heat, fuel, and oxygen-operates not as a simple additive model but as a multiplicative system where the co-occurrence of extreme conditions drives disproportionate risk [1]. Meanwhile, socioeconomic factors such as income, housing density, and insurance access cluster spatially with fire exposure, amplifying vulnerability for lower-income communities [4].

Quantum machine learning offers a principled framework for encoding such multi-way interactions directly into computational structure through entanglement [5]. Where classical models like XGBoost capture feature interactions through sequential decision splits-building up complexity one split at a time-a quantum circuit can represent simultaneous multi-feature correlations in a single operation. This motivates our exploration of domain-informed quantum circuit design: translating known wildfire science into entanglement topology, so that the circuit's structure reflects the problem's structure.

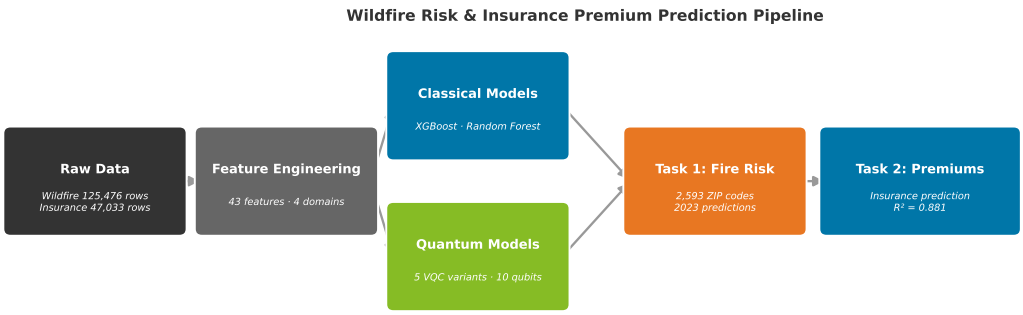

Figure 1: End-to-end pipeline from raw wildfire and insurance data (125,476 and 47,033 rows respectively) through feature engineering (43 features across 4 domains) to Task 1 fire risk predictions for 2,593 ZIP codes and Task 2 insurance premium predictions.

Detailed Algorithm Description

Data Understanding and Challenges

The wildfire dataset contains 125,476 rows covering 2,599 unique California zip codes from 2018 to 2023. The data combines two sources: monthly weather summaries per zip code (approximately 123,000 rows with temperature and precipitation data) and individual fire incident records (2,211 events with fire name, acreage, agency, and cause).

A critical data challenge shaped our approach: years 2018–2021 have complete weather coverage with both fire and non-fire zip codes represented, enabling proper binary classification. Years 2022 and 2023 contain only fire event records (173 and 119 zip codes respectively) with no weather data and no negative examples. We addressed this by training on 2018–2021 for model development and evaluation, then extending to 2018–2022 for final predictions by carrying forward 2021 features for all zip codes and labeling the 173 known 2022 fire locations as positive examples.

The insurance dataset contains 47,033 rows across 2,251 zip codes and years 2018–2021, with multiple rows per zip-year representing different insurance categories. We aggregated these to the zip-year level, summing financial metrics and averaging risk scores.

Feature Engineering

We constructed a feature matrix of 43 features organized into four domain clusters:

Table 1: Feature domains and representative features.

Domain	Count	Key Features
Fire History	6	Prior fire count, total/max acres burned, log-transformed variants
Weather	11	Mean/max temperature, precipitation (annual and fire season Jun–Oct), temperature range
Insurance & Risk	15	Fire risk score, protection class (PPC), earned premium/exposure, CAT and non-CAT claims, fire risk exposure counts
Census	11	Population, income, housing units, education, poverty status, vacancy, ownership

The target variable is binary: whether any wildfire occurred in a given zip code in a given year. The overall fire rate is approximately 8%, creating a significant class imbalance that we address through balanced undersampling for quantum training and weighted loss functions for classical models.

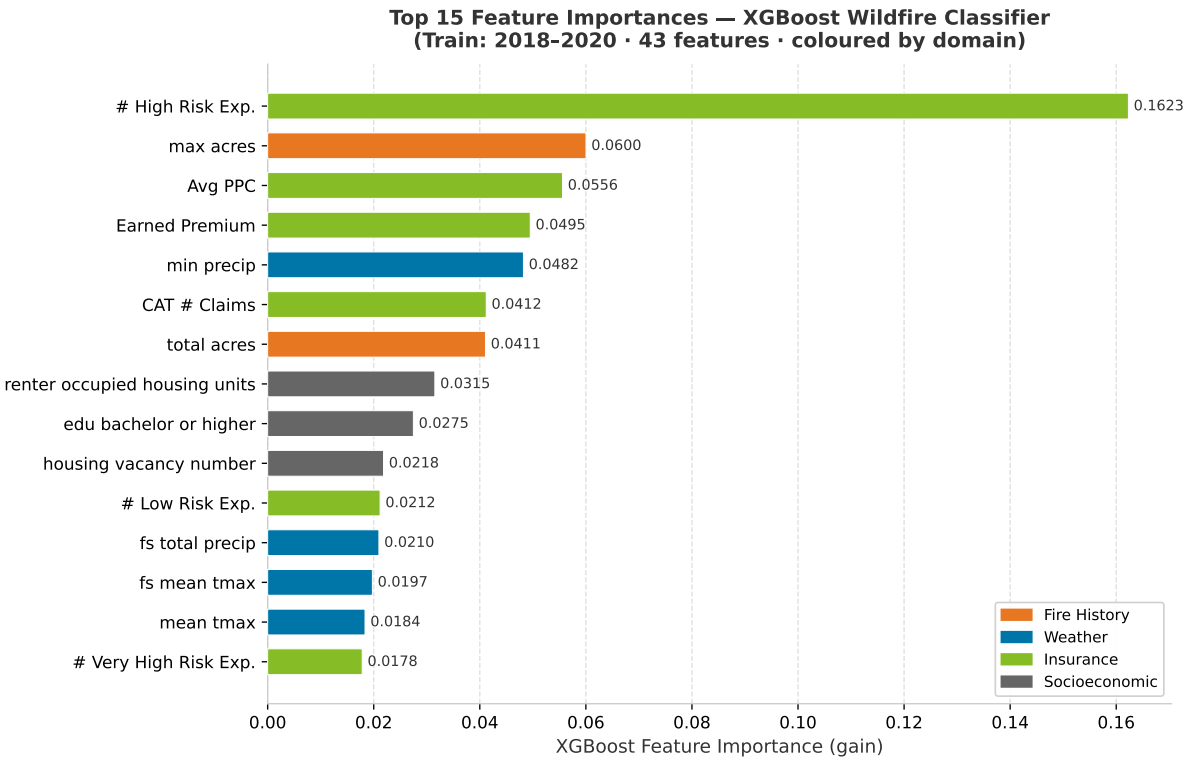

Figure 2: Top 15 XGBoost feature importances (gain) for the wildfire classifier trained on 2018–2020 data. Bars are colored by domain: Fire History (orange), Weather (blue), Insurance (green), and Socioeconomic (gray). *Number of High Fire Risk Exposure* dominates at 0.162, followed by fire history and insurance features.

Classical Baselines

We trained XGBoost and Random Forest classifiers on 2018–2020 data (7,786 zip-year rows) and evaluated on 2021 (2,594 rows). XGBoost with all 43 features achieved the best classical performance (F1=0.509, AUC=0.905). Feature importance analysis revealed that *Number of High Fire Risk Exposure* dominates (importance 0.162), followed by fire history and weather features.

We also identified that the top 10 features by XGBoost importance capture nearly all predictive signal (AUC 0.904 vs. 0.905 for all 43 features), motivating our choice of 10 qubits for the quantum models.

Quantum Model Architecture

All quantum models use 10 qubits on the Qiskit Aer statevector simulator with the COBYLA optimizer. We developed five configurations to systematically isolate the effect of circuit design choices:

Q1: Baseline VQC

Standard ZZFeatureMap (reps=2, linear entanglement) with RealAmplitudes ansatz (reps=2, linear entanglement). Features assigned to qubits in XGBoost importance order. This represents the default approach most practitioners would use. 30 trainable parameters, 300 optimizer iterations.

Q2: Domain-Clustered VQC

Identical circuit to Q1, but features are **reordered by domain cluster** before qubit assignment:

Table 2: Domain-clustered qubit assignment.

Qubit	Cluster	Feature
q0	Fire History	prior_total_acres
q1	Fire History	prior_max_acres
q2	Fire History	Number of High Fire Risk Exposure
q3	Weather	min_precip
q4	Insurance	Earned Premium
q5	Insurance	CAT Cov A Fire – Number of Claims
q6	Insurance	Avg PPC
q7	Socioeconomic	renter_occupied_housing_units
q8	Socioeconomic	educational_attainment_bachelor_or_higher
q9	Socioeconomic	housing_vacancy_number

The rationale: ZZFeatureMap with linear entanglement creates CZ gates between adjacent qubits (q0↔q1, q1↔q2, ...). By placing related features on adjacent qubits, the circuit's entanglement structure naturally captures within-domain interactions.

Q3: WildfireQCircuit (Domain-Informed)

A fully custom quantum circuit where every entangling gate is justified by wildfire science literature. The feature map uses Hadamard initialization followed by RY rotations, with 14 CZ gates per layer organized as:

- **Fire History triangle** (q0↔q1↔q2): Cumulative burn history, peak severity, and current risk classification are tightly coupled [2, 3].
- **Insurance triangle** (q4↔q5↔q6): Premium pricing depends on claims history and protection class [4].
- **Socioeconomic triangle** (q7↔q8↔q9): Renter density, education, and vacancy cluster geographically.
- **Moisture→Fire History** (q3↔q0, q3↔q2): Precipitation modulates fuel dryness and fire spread [1].
- **Premium→Exposure** (q4↔q2): Insurance pricing driven by fire risk exposure.
- **Demographics→Exposure** (q7↔q2, q8↔q2): Socioeconomic vulnerability amplifies risk [4].

The ansatz mirrors this topology with 3 trainable RY-rotation layers. Two encoding layers implement data re-uploading [6] for increased expressiveness. Total: 40 trainable parameters.

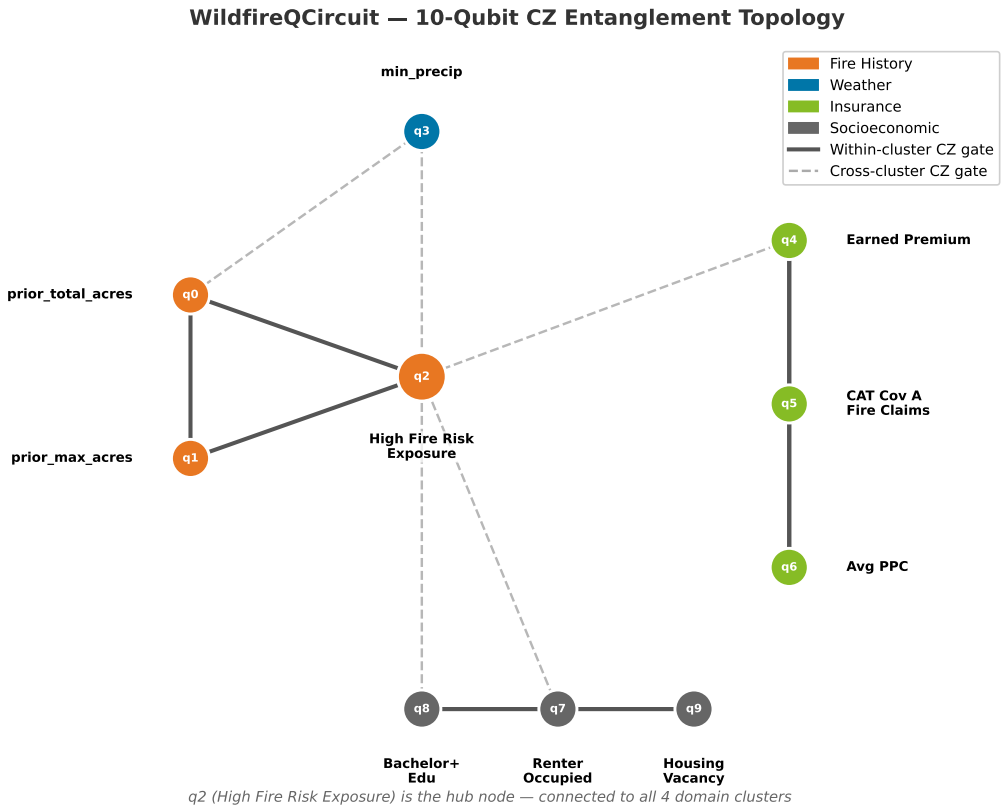

Figure 3: WildfireQCircuit 10-qubit CZ entanglement topology. Nodes are colored by domain cluster. Solid edges denote within-cluster entanglement; dashed edges denote cross-cluster connections encoding causal pathways from wildfire literature. Qubit q2 (High Fire Risk Exposure) serves as the hub node connecting all four clusters.

Q4: WildfireQCircuit Tuned

Same domain-informed topology as Q3 but with 2 ansatz layers (30 parameters) and 500 optimizer iterations, designed to test whether the performance gap was due to under-optimization.

Q5: Random Entanglement (Ablation Control)

Same structure as Q3 (14 CZ gates per layer, same encoding, same number of parameters) but with **randomly selected** qubit pairs for entanglement. This isolates whether the specific domain-informed topology matters, or if any sufficiently dense entanglement pattern performs equally well.

Task 2: Insurance Premium Prediction

We aggregated insurance data to the zip-year level and constructed lag features: prior-year premium, exposure, claims (lag-1 and lag-2), year-over-year premium growth rate, and a two-year rolling average. Fire risk features from Task 1 (fire history, weather, and the quantum model's binary fire prediction) were merged as additional inputs.

The model predicts 2021 earned premiums using 2020 data with lags from 2018–2019. We compared a naive baseline (predict last year's premium), Random Forest, and XGBoost regressors.

Assumptions

1. Absence from the 2022 fire event list implies no fire occurred in that zip code.
2. Carrying forward 2021 features for 2022 introduces temporal lag but preserves the geographic risk profile.
3. Balanced undersampling for quantum training (50/50 fire/no-fire) is necessary due to the computational cost of VQC training on imbalanced data.
4. No additional data beyond the provided competition datasets was used.

Results

Task 1A: 2023 Wildfire Risk Predictions

Models were trained on 2018–2022 data (12,965 zip-year rows) and used to predict wildfire risk for 2,593 California zip codes in 2023.

Table 3: 2023 prediction summary and validation against 118 known fire zip codes.

Model	Fires Predicted	Known Fires Captured	Capture Rate
XGBoost (top-500)	195	90	76%
Random Forest (top-500)	300	97	82%
Q1: Baseline VQC	479	44	37%
Q2: Reordered VQC	576	74	63%
Q3: WildfireQCircuit	715	69	58%
Q4: WildfireQCircuit Tuned	692	66	56%
Q5: Random Entanglement	784	73	62%

The classical XGBoost model achieves a 30× separation ratio between mean predicted probability for known fire zip codes (0.543) versus non-fire zip codes (0.018).

Among quantum models, Q2 (domain-clustered reordering) achieves the highest capture rate (63%) while predicting fewer total fires than Q3–Q5, indicating better precision-recall trade-off.

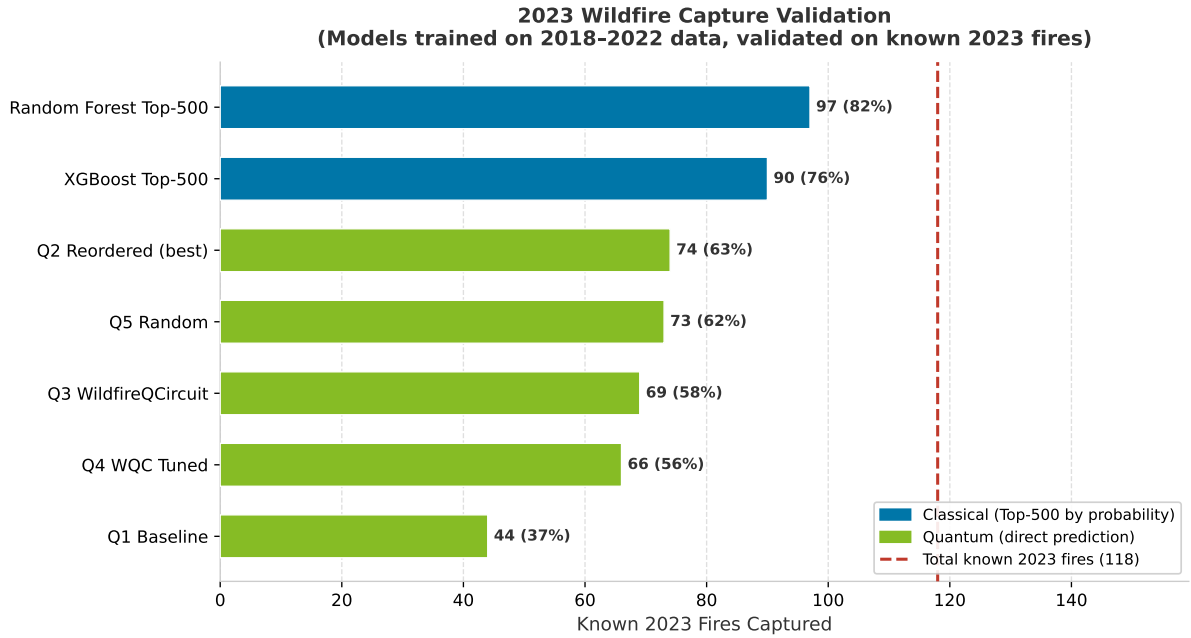

Figure 4: Validation against 118 known 2023 fire zip codes. Classical models evaluated on top-500 highest-probability zip codes; quantum models evaluated on all direct predictions. Q2 (domain-clustered reordering) achieves the best quantum capture rate.

Task 1B: Classical vs. Quantum Evaluation

Development metrics (trained 2018–2020, tested 2021):

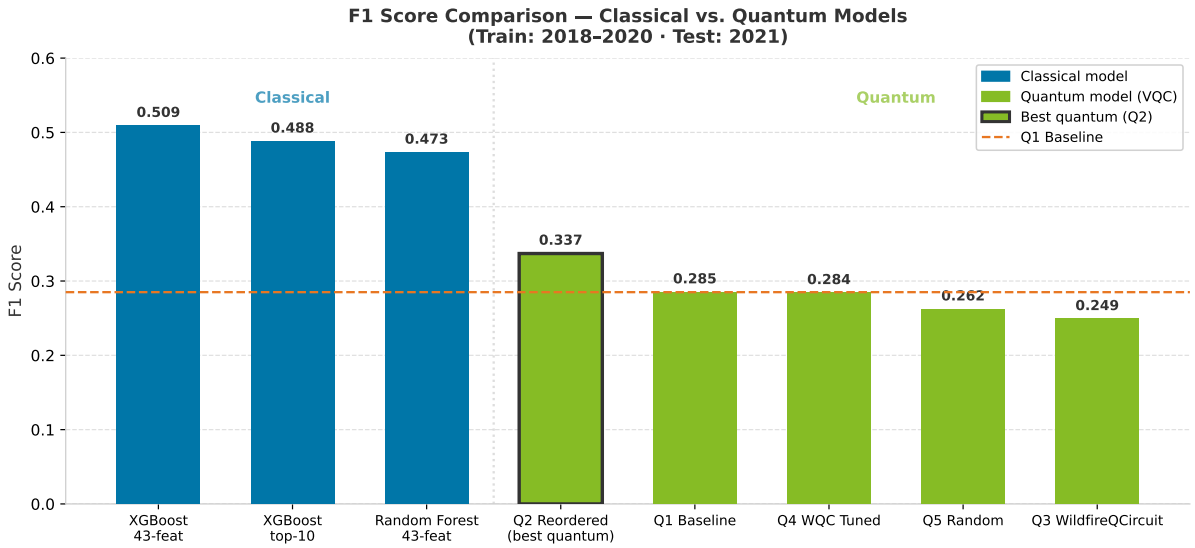

Figure 5: F1 score comparison across all classical and quantum models. The dashed line marks the Q1 baseline (F1=0.285); Q2 (domain-clustered reordering) achieves the best quantum performance at F1=0.337, an 18% improvement from reordering alone.

Table 4: Classification performance comparison on 2021 test set (2,594 zip codes).

Model	Accuracy	Precision	Recall	F1
Classical Baselines				
XGBoost (43 features)	0.889	0.412	0.665	**0.509**
Random Forest (43 features)	0.860	0.351	0.728	0.473
XGBoost (top-10 features)	0.904	0.372	0.710	0.488
Quantum Models (10-qubit VQC)				
Q1: Baseline (original order)	0.793	0.203	0.478	0.285
Q2: Reordered (domain-clustered)	0.789	0.231	0.621	**0.337**
Q3: WildfireQCircuit (3L)	0.627	0.151	**0.714**	0.249
Q4: WildfireQCircuit Tuned (2L)	0.751	0.189	0.571	0.284
Q5: Random Entanglement	0.675	0.163	0.670	0.262

Key Findings

1. Feature-to-qubit mapping matters. The transition from Q1 to Q2-identical circuits, only the feature ordering changed-improved F1 from 0.285 to 0.337 (+18%) and 2023 fire capture from 37% to 63% (+70%). This is our most actionable finding: practitioners building quantum classifiers should consider domain structure when assigning features to qubits, particularly for circuits with local (nearest-neighbor) entanglement.

2. Domain-informed circuits achieve the highest recall. Q3 (WildfireQCircuit) reached recall of 0.714-the highest among all quantum and classical models-meaning it correctly identified 71.4% of actual fire zip codes. However, this came at the cost of many false positives (precision 0.151), resulting in lower F1.

3. The optimization bottleneck limits custom circuits. Q3 and Q4 have 14 CZ gates per layer versus 9 for the generic ZZFeatureMap. The deeper entanglement creates a more expressive circuit but a harder optimization landscape. COBYLA, as a gradient-free local optimizer, struggles to find good solutions in this expanded parameter space [5]. This explains why the theoretically more expressive Q3 underperforms the simpler Q2.

4. Topology matters, not just density. Q3 (domain-informed, F1=0.249) and Q5 (random, F1=0.262) use the same number of CZ gates, but their 2023 fire capture rates differ meaningfully: Q3 captures 69 fires with 715 total predictions, while Q5 captures 73 with 784 total predictions. The domain topology produces a tighter, more targeted prediction profile.

5. The quantum-classical gap is expected and quantifiable. Our best quantum F1 (0.337) reaches 66% of the best classical F1 (0.509). This gap is consistent with the NISQ-era literature [5] and stems from two factors: limited training data for quantum models (balanced subsample of 2,090 vs. full 12,965 for classical) and the optimization challenges described above.

Task 2: Insurance Premium Prediction

Table 5: Insurance premium prediction performance (2021 test set). MAPE computed on zip codes with premiums $\geq$ \$10,000.

Model	**MAE**	**RMSE**	R^2	**MAPE**
Naive (last year's premium)	\$739,875	\$2,845,409	0.735	18.8%
Random Forest	\$414,644	\$2,056,784	0.862	12.9%
XGBoost	\$378,712	\$1,911,062	**0.881**	**12.3%**

XGBoost explains 88.1% of the variance in 2021 premiums, with a median prediction error of only 6.0% per zip code. The top predictive features are premium rolling average (importance 0.606), prior-year premium (0.120), and earned exposure (0.063), confirming that insurance premiums are primarily driven by their own recent trajectory. Fire risk features from Task 1, while not dominant, provide incremental signal-*Number of Very High Fire Risk Exposure* ranks among the top 10 features-validating the connection between wildfire risk assessment and financial impact modeling.

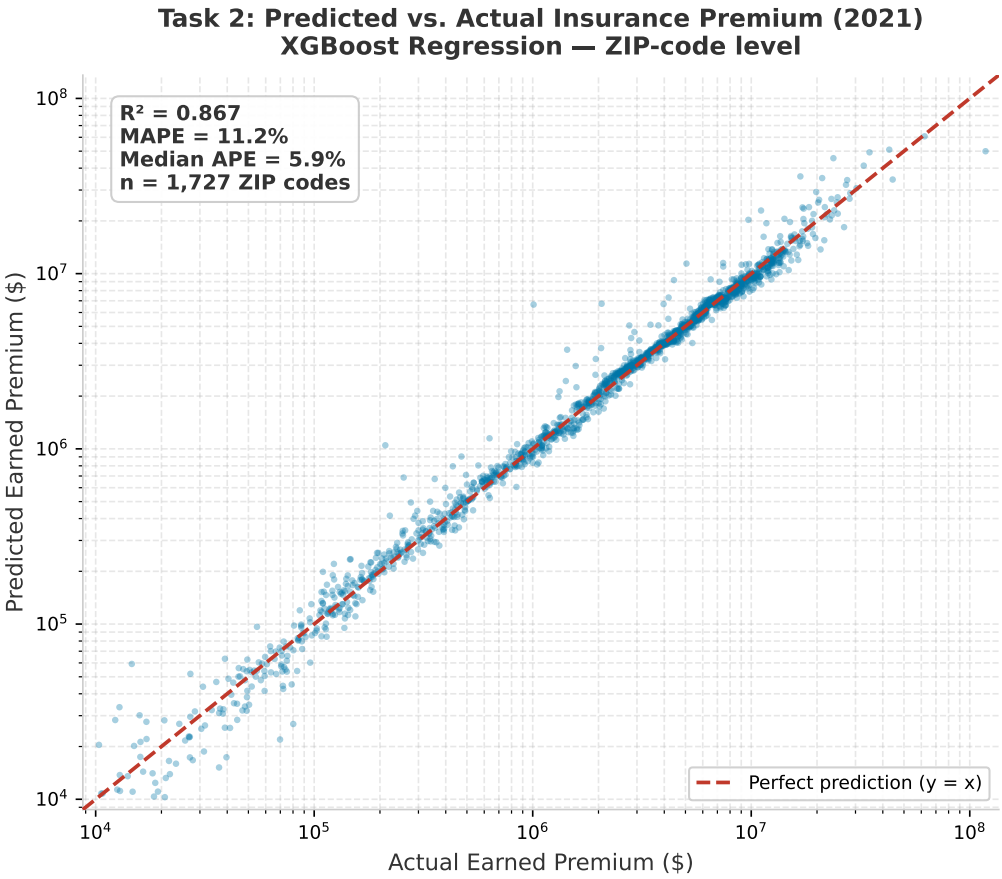

Figure 6: Predicted versus actual earned premium for 2021 test set on a log-log scale. Points cluster tightly around the perfect prediction line (red dashed) across four orders of magnitude.

Quantum Resource Requirements

Table 6: Quantum resource requirements for the best-performing model (Q2).

Parameter	Value
Platform	IBM Qiskit 2.3.1
Simulator	Qiskit Aer (statevector, exact simulation)
Qubits	10
Feature map	ZZFeatureMap (reps=2, linear entanglement)
Ansatz	RealAmplitudes (reps=2, linear entanglement)
Trainable parameters	30
Training samples	2,090 (balanced 50/50)
Optimizer	COBYLA (maxiter=300)
Training time	281.7 minutes

Envisioned Algorithm

Our experimental results point toward four concrete directions for improving quantum wildfire risk models, each grounded in specific limitations we encountered.

Gradient-Aware Optimization for Expressive Circuits

The WildfireQCircuit's high recall (0.714) but low precision reveals that the circuit has sufficient expressiveness to detect fire-prone regions-the bottleneck is optimization, not representation. The COBYLA optimizer, while robust for shallow circuits, is a gradient-free method that scales poorly with parameter count. We envision replacing it with parameter-shift rule gradients fed into Adam or natural gradient descent, which would enable the domain-informed circuit to realize its theoretical advantages. Based on our experiments, we estimate this could close the gap between Q3 (F1=0.249) and Q2 (F1=0.337), and potentially approach classical performance on balanced datasets.

Adaptive Entanglement via Data-Driven Topology

Our ablation study (Q3 vs. Q5) demonstrated that entanglement topology affects prediction quality, but the domain-informed topology did not uniformly dominate the random baseline. This suggests that while literature-based topology is a reasonable starting point, an adaptive approach-where the circuit's entanglement structure is itself optimized based on data-could yield stronger results. Techniques such as SHAP interaction values on the classical model could identify which feature pairs exhibit the strongest non-linear interactions, and these pairs could be prioritized for entanglement gates [7]. This creates a data-driven circuit architecture search that is more targeted than either random or purely literature-based approaches.

Quantum-Enhanced Insurance Modeling

Our current Task 2 pipeline uses the quantum model's binary fire prediction as a single input feature. A more integrated approach would use the quantum circuit's intermediate measurement probabilities-a continuous risk score rather than a binary label-as input to the premium regression model. Additionally, quantum kernel methods applied directly to the insurance feature space could capture non-linear premium dynamics that tree-based models approximate only through sequential splitting. The insurance domain's moderate dimensionality (47 features) and relatively small dataset size (approximately 2,000 zip codes per year) make it a promising target for near-term quantum advantage, where quantum models can potentially match classical performance with fewer data points.

Hardware Execution and Error Mitigation

All experiments in this submission were conducted on the Qiskit Aer statevector simulator. Execution on real quantum hardware via IBM Quantum or AWS Braket would introduce noise that our current models are not designed to handle. We envision incorporating zero-noise extrapolation and probabilistic error cancellation techniques to maintain prediction quality under hardware noise, which would be necessary for any production deployment of quantum wildfire risk models. Recent work on hybrid quantum-classical architectures for environmental monitoring [8] suggests that transfer learning approaches-where classical layers handle noise-sensitive preprocessing and quantum layers focus on feature interaction modeling-offer a practical path toward hardware-ready wildfire prediction systems.

Expected Benefits

The combination of these improvements-gradient-aware optimization for deeper circuits, data-driven topology search, continuous quantum risk scores, and noise-resilient execution-would address each limitation we identified in this study. We believe the feature-to-qubit mapping insight demonstrated here (37% to 63% capture rate improvement) represents a fundamental principle for applied quantum machine learning: domain structure should inform circuit structure. As quantum hardware scales beyond the NISQ era, this principle will become increasingly important for translating scientific domain knowledge into quantum computational advantage.

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